	S.S.V.P.S's B. S. Deore Polytechnic, Dhule **Department of Computer Engineering** *Academic Year : 2025 – 2026*

Presentation on Capstone Project Execution (CPE–316004)

Topic	Site's Material Management System	
Presented by	1. Tanushree Parag Ahire	
	2. Yashika Paresh Chaudhari	
	3. Snehal Tukaram Deore	
	4. Antara Ramesh Pahade	
Guided by	Ms. Shraddha.S.Hajare	
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Abstract

The **Site Material Management System** is a web-based solution developed to efficiently manage construction materials across multiple sites. It enables supervisors to view available materials and raise requests, while administrators can manage inventory, approve requests, and monitor system activities through a centralized dashboard.

The system includes features like secure login, real-time stock tracking, low stock alerts, and usage history, ensuring better transparency and control. By reducing manual work and improving accuracy, the system enhances decision-making, prevents material shortages, and supports smooth construction operations.

Content

1. Introduction (background of the industry or used based problem / task)
2. Literature Survey
3. Scope of the project
4. Methodology
5. Details of designs, working and processes
6. Results and applications
7. Conclusion and future scope
8. References and Bibliography

1. Introduction

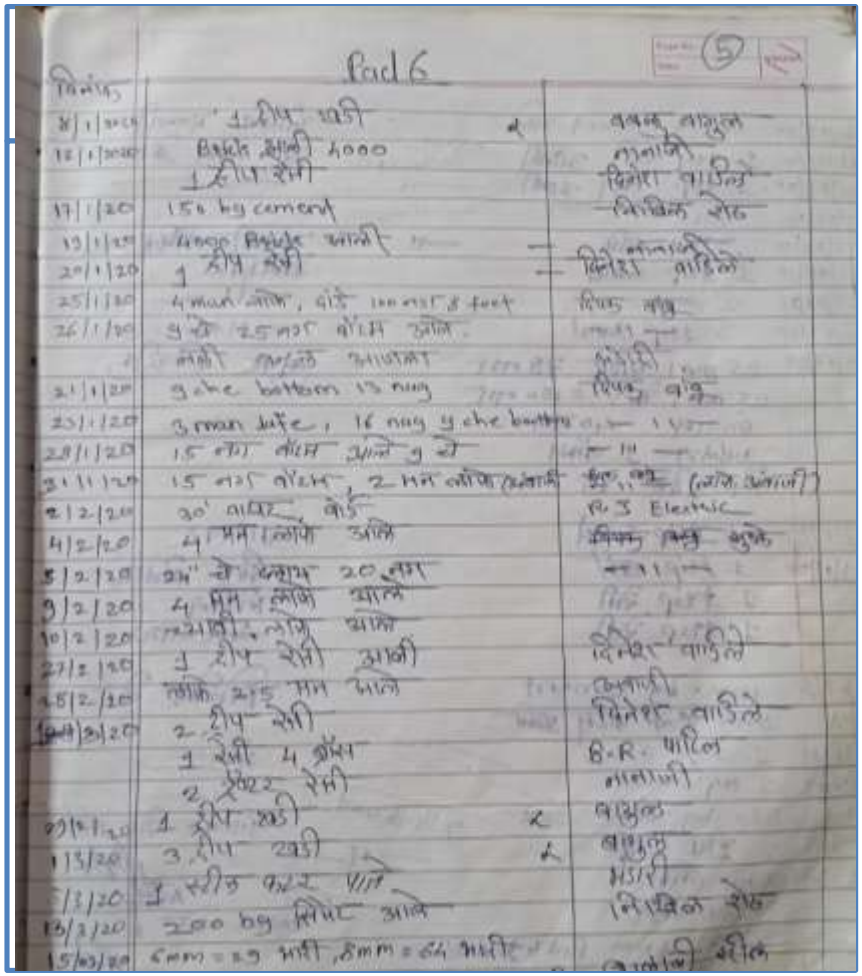
The **Site Material Management System** is a web-based application designed to simplify and automate the management of construction materials across multiple sites. In traditional methods, material tracking and request handling are done manually, which often leads to errors, delays, and lack of transparency.

This system provides a digital platform where supervisors can request materials and monitor availability, while administrators can manage inventory, approve requests, and track usage efficiently. The system improves coordination, ensures accurate record-keeping, and enhances overall productivity in construction projects.

2. Literature Review

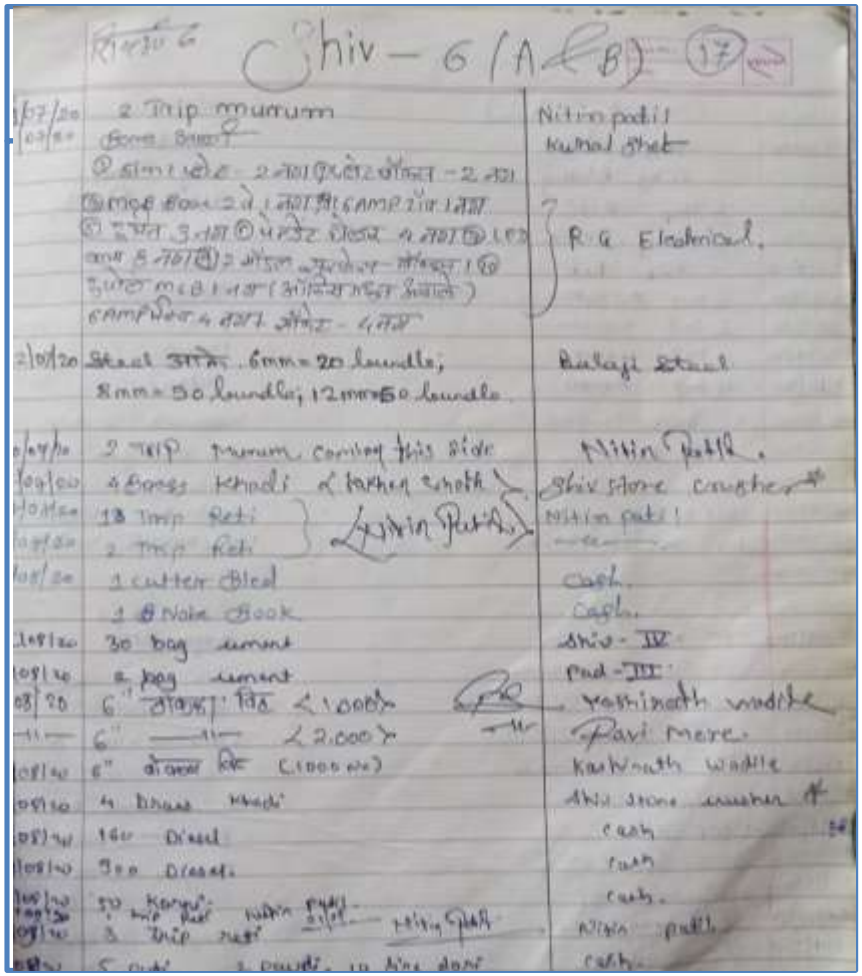
- Existing systems use manual or semi-digital methods for material management
- Manual processes lead to errors, delays, and lack of transparency
- Some systems provide inventory tracking but lack real-time updates
- Limited integration between material requests and approval processes
- Difficulty in tracking material usage across multiple sites
- Lack of proper alert systems for low stock materials
- Proposed system overcomes these limitations with automation and real-time monitoring
- Provides centralized control, improved accuracy, and better decision-making

2.1 Study of existing system



- The system maintains material records manually in a register/notebook
- The notebook contains details like date, material name, quantity, and usage
- Supervisor and supplier details are written for tracking purposes
- Total material usage and stock are calculated manually
- Records are maintained in handwritten format, making management difficult
- Manual calculations and entries increase chances of errors and data loss

2.1 Study of existing system



- The supervisor maintains material and expense records manually in a notebook
- It includes details like material type, quantity, supplier name, and cost
- Daily material usage such as cement, steel, diesel, etc., is recorded
- Payments and expenses are noted along with material entries
- All calculations and records are done manually
- Managing and tracking records becomes difficult and time-consuming

2.2 Problem Statement

Existing System / Manual Process

The existing system of material management at construction sites is manual and paper-based, where supervisors record all material details in notebooks. This approach leads to errors, data inconsistency, and difficulty in maintaining accurate records. Tracking real-time material availability becomes challenging, and searching past data is time-consuming. There is no proper system for managing material requests and approvals, which affects coordination between supervisors and admin. Additionally, the absence of low stock alerts can cause delays in construction work. Overall, the manual system lacks efficiency, transparency, and reliability.

2.2 Problem Statement

Problems and Limitations

- Manual record keeping leads to human errors
- Difficult to track and update material records
- No real-time visibility of stock availability
- Time-consuming to search past data
- No alerts for low stock, causing delays

Need for Improvement

- To replace manual record keeping with a digital system for better accuracy and efficiency
- To enable real-time tracking and alerts for effective material management

2.3 Problem Definition

The current system of managing construction materials is manual and inefficient, where supervisors maintain records in notebooks. This leads to errors, difficulty in tracking material availability, and delays in decision-making. There is no proper system for handling material requests, approvals, or stock monitoring. As a result, the process lacks transparency, accuracy, and real-time control, affecting overall project efficiency.

3. Scope of the project

- Develop a web-based system for managing construction materials
- Allow supervisors to request and track materials easily
- Enable admin to manage inventory and approve requests
- Provide real-time stock monitoring and low stock alerts
- Maintain records of material usage and history
- Improve efficiency, accuracy, and transparency in material management

3.1 User Requirements

- **Supervisor:**
- Secure login authentication
- View available materials
- Create material requests
- Track request status
- Select and manage assigned site
- View personal request history

3.1 User Requirements

- **Admin:**
- Secure admin login
- Add, update, and delete materials
- Approve or reject requests
- Monitor stock and low stock alerts
- Manage supervisors and site allocation
- View reports and usage history

3.2 System Requirements

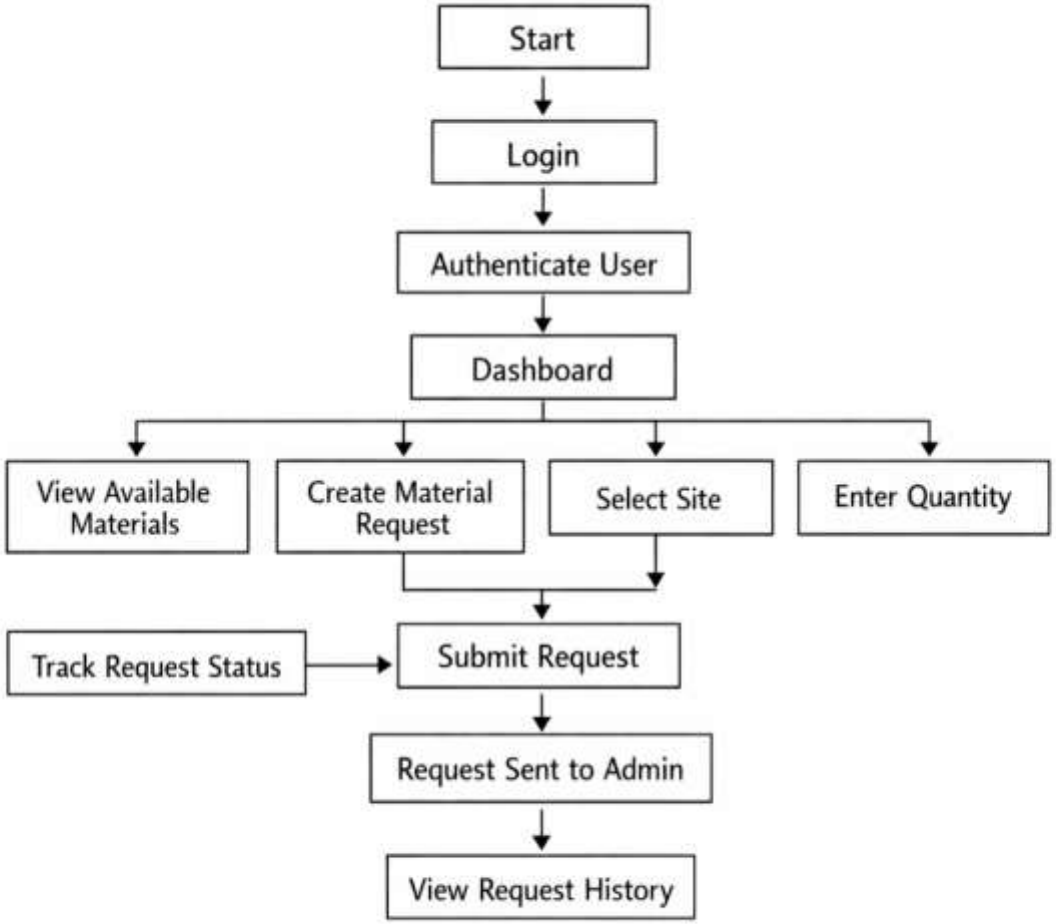
- Programming Languages - JavaScript
- Frontend - React JS, HTML, CSS
- Backend - Node.js, Express.js
- Database - MongoDB
- Development Tools - VS Code
- Operating System – Windows 10

4. Methodology

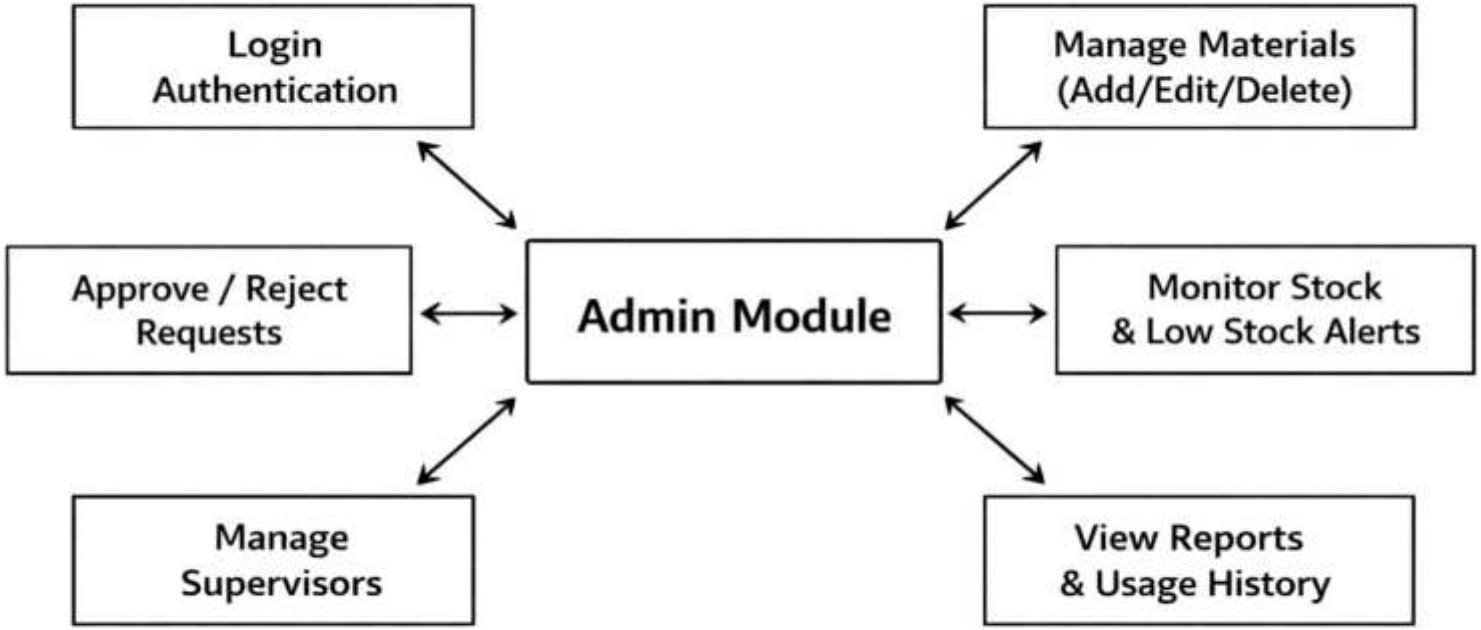
Modules

- Supervisor Module
- Admin Module
- Material Management Module
- Request Management Module
- Inventory & Stock Module
- Reports & History Module

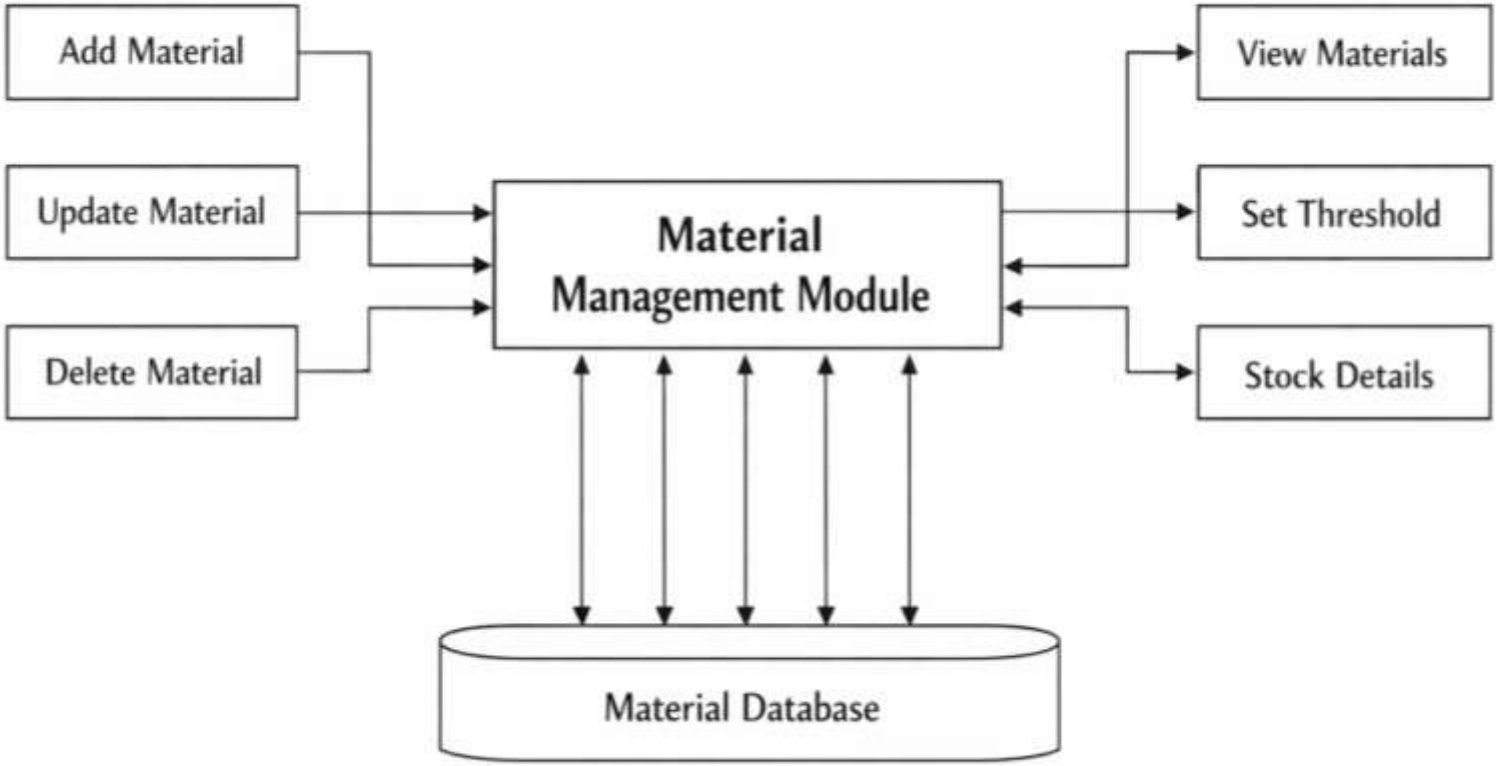
4. 1 Supervisor Module



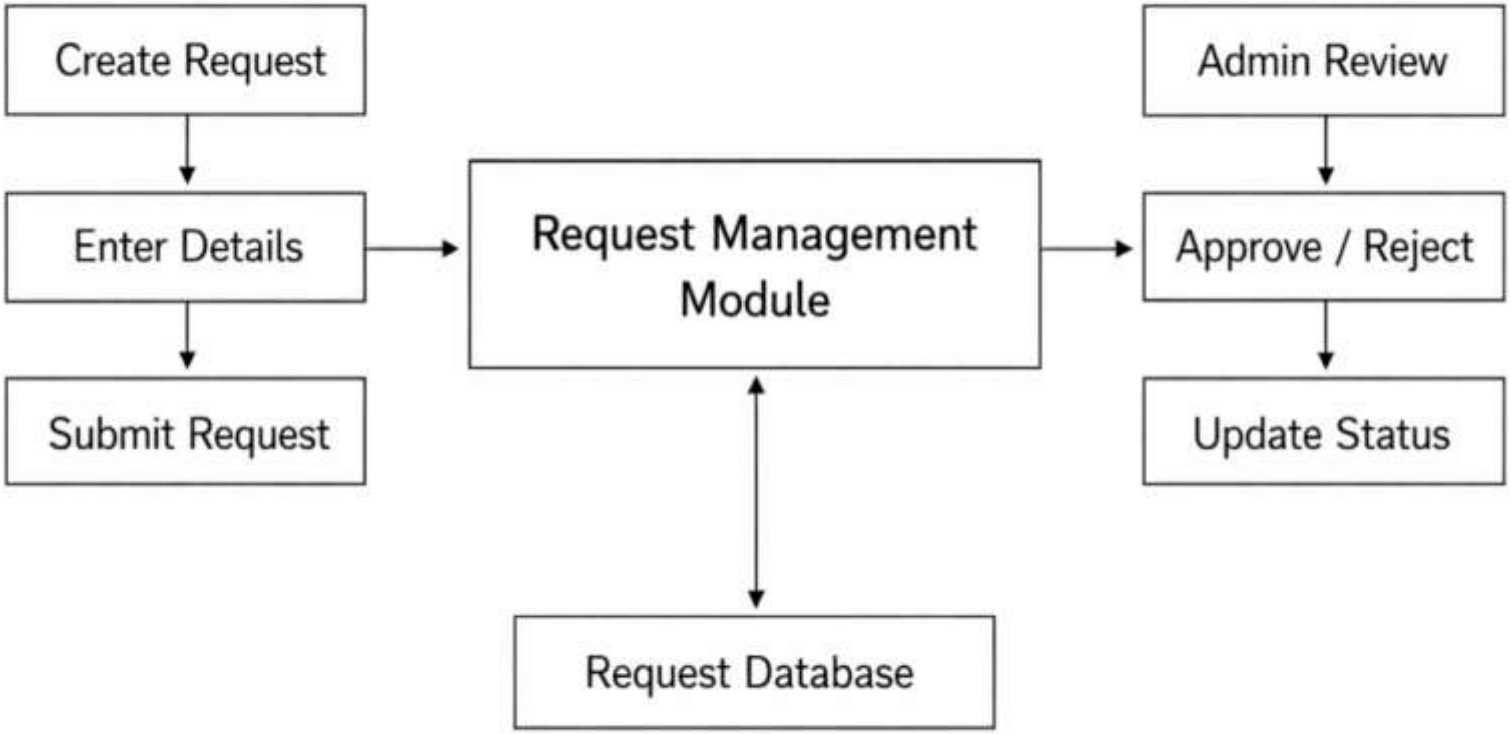
4. 2 Admin Module



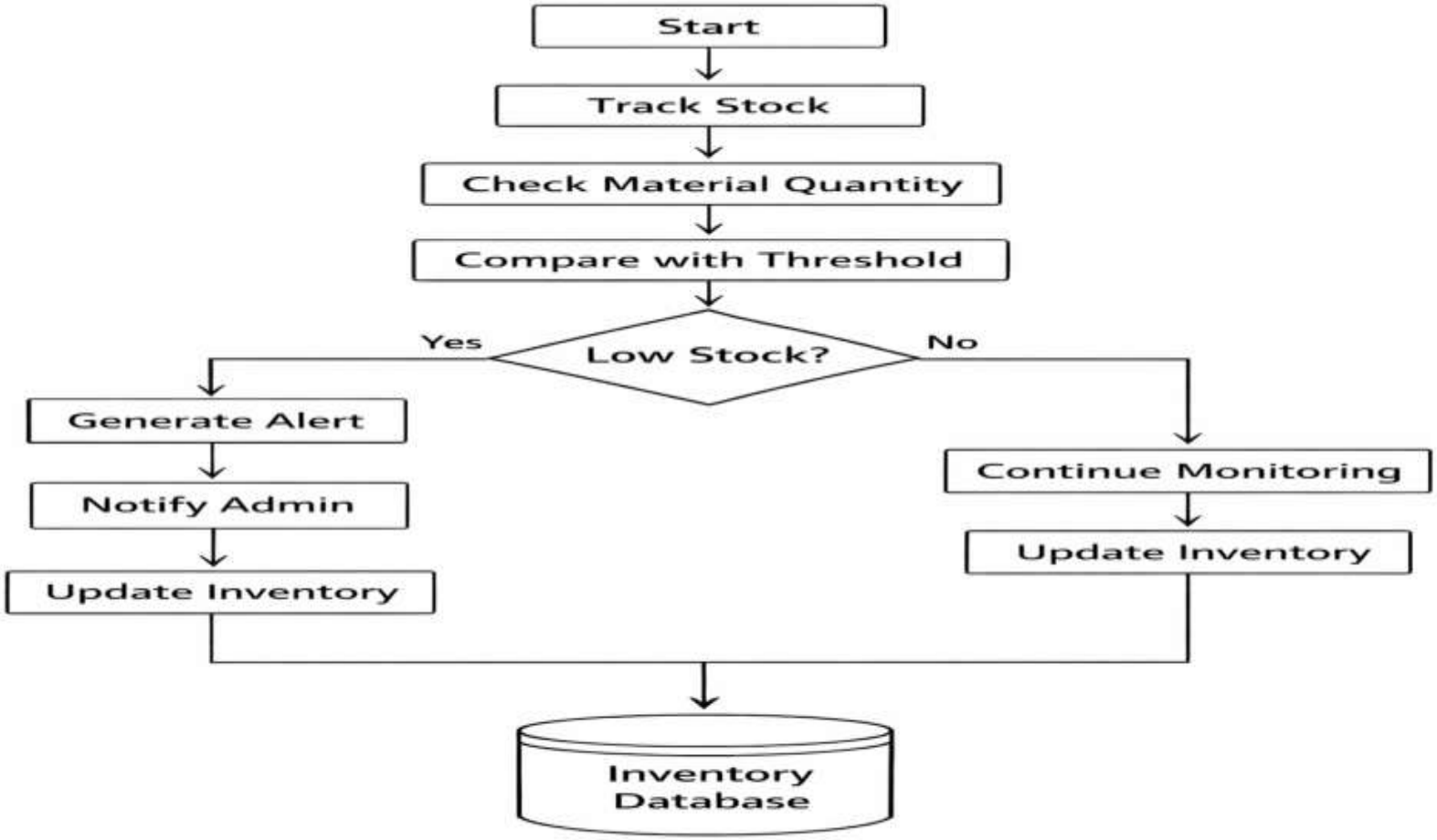
4. 3 Material Management Module



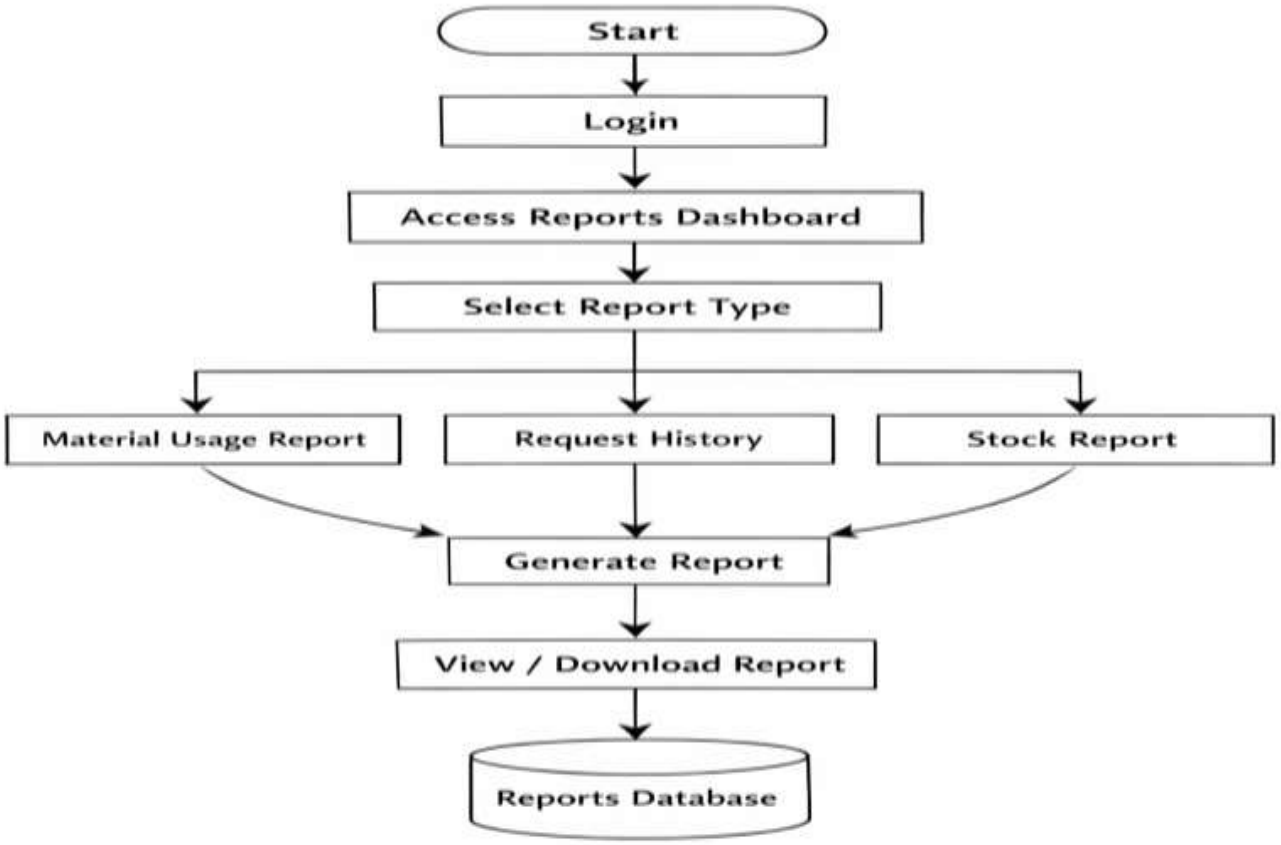
4.4 Request Management Module



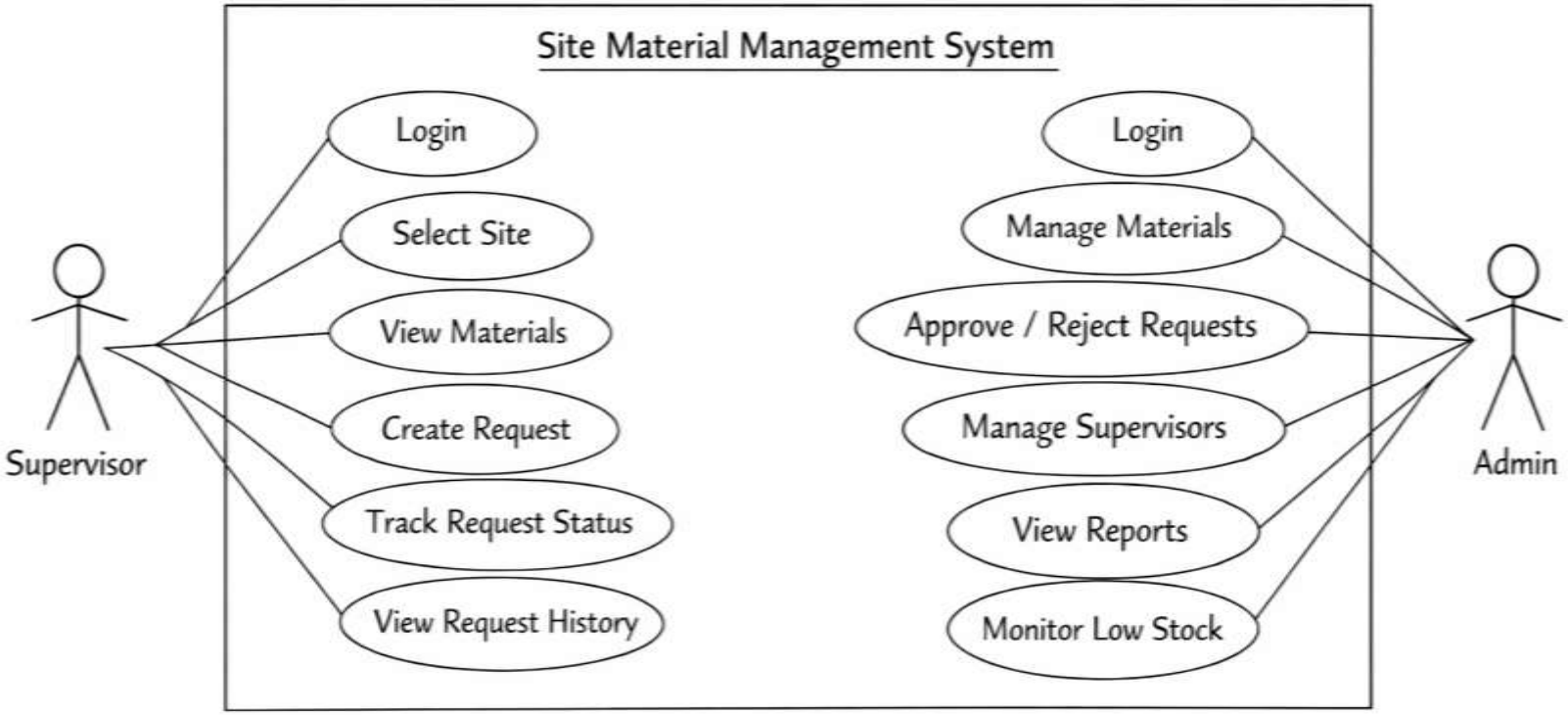
4.5 Inventory & Stock Module



4.6 Reports & History Module



4. 4 Use Case Diagram



4. 5 Tools and Technology

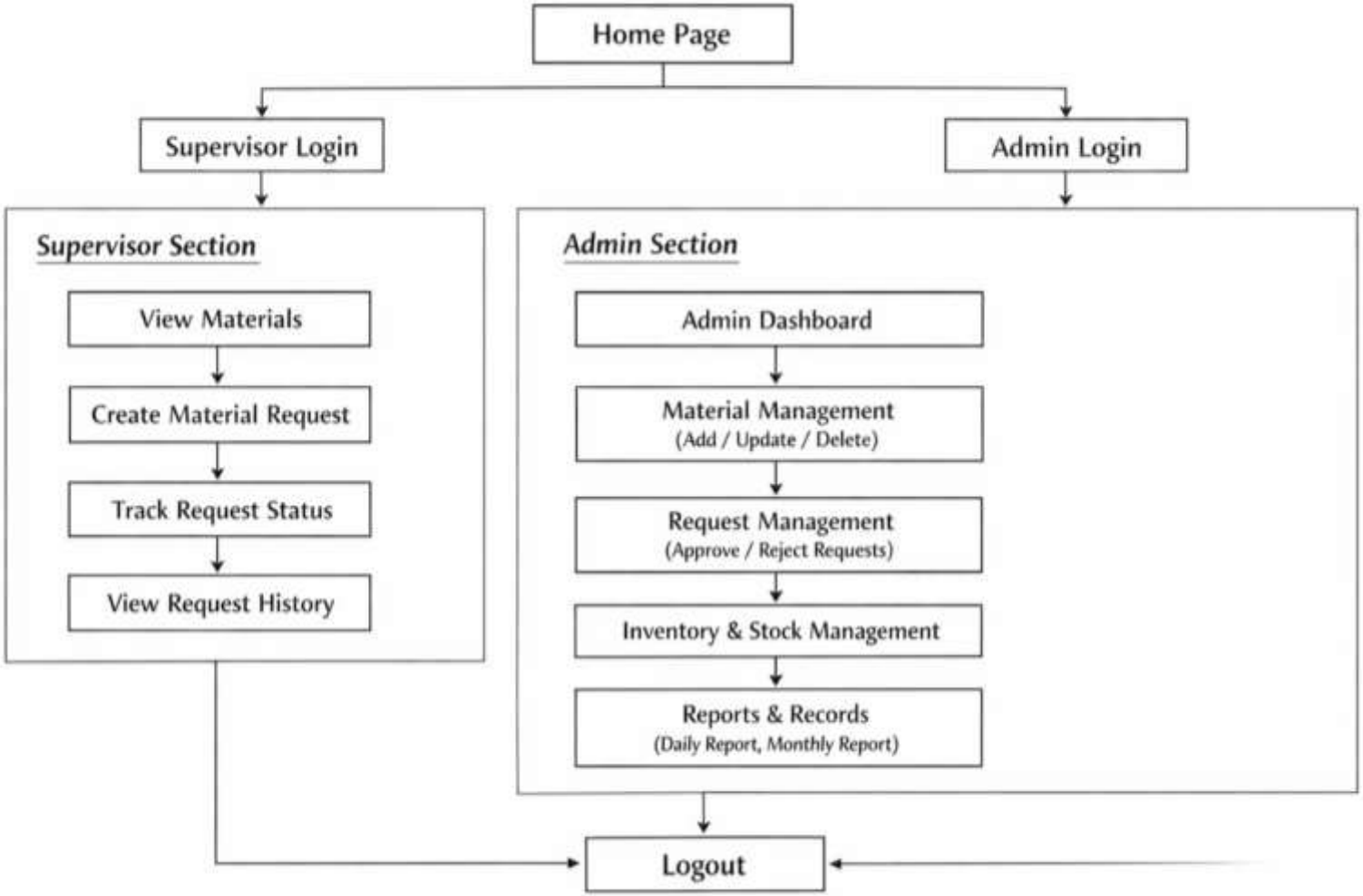
- Programming Languages - JavaScript
- Frontend - React JS, HTML, CSS
- Backend - Node.js, Express.js
- Database – MongoDB
- Development Tools - VS Code
- Operating System – Windows 10

5. Details of designs, working & processes

Flow of Operations

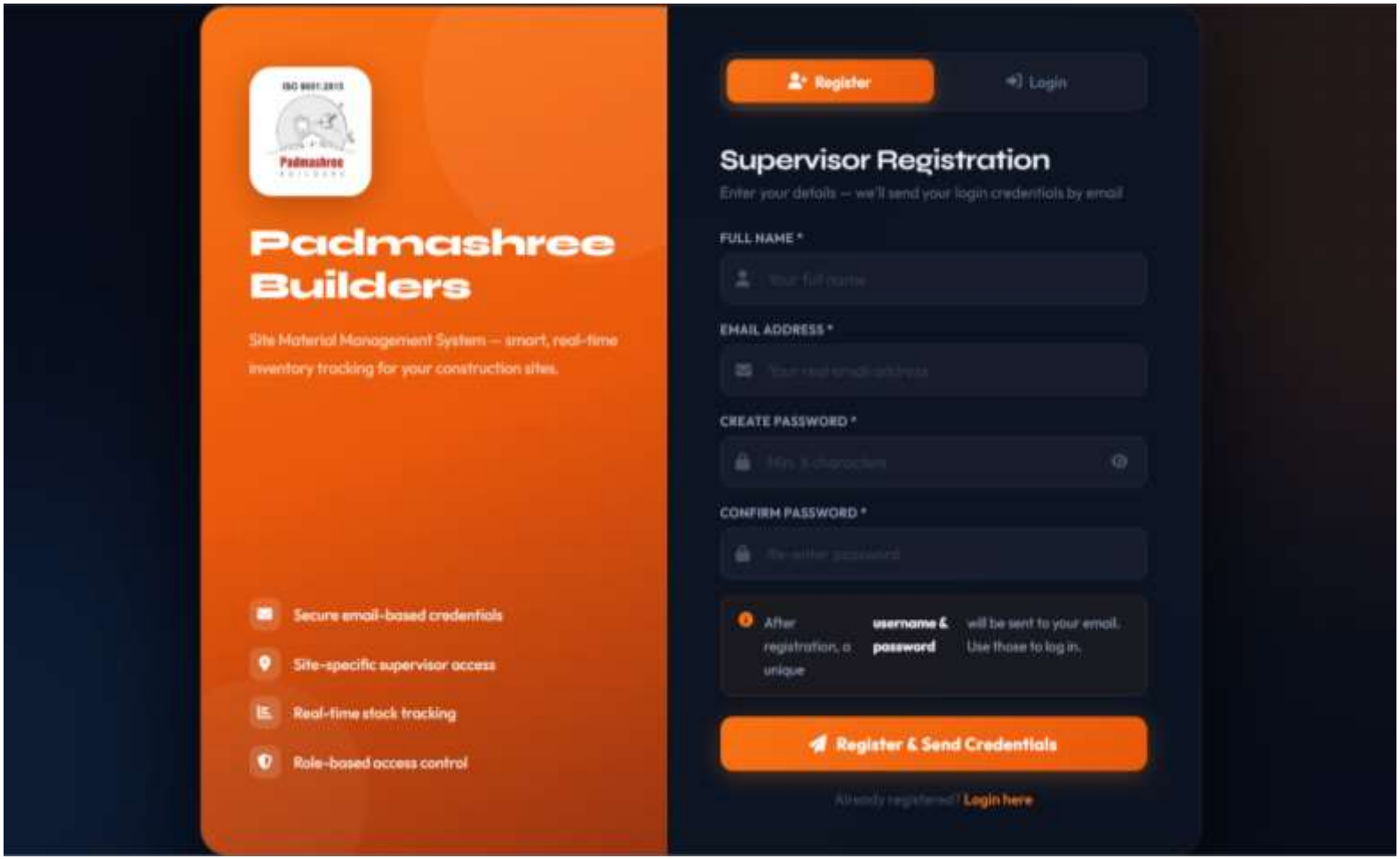
1. Supervisor logs into the system
2. Supervisor views available materials
3. Supervisor places a material request
4. Request is stored in the system
5. Admin views the material request
6. Admin approves or rejects the request
7. Material stock is updated and report is generated

5. Details of designs, working & processes

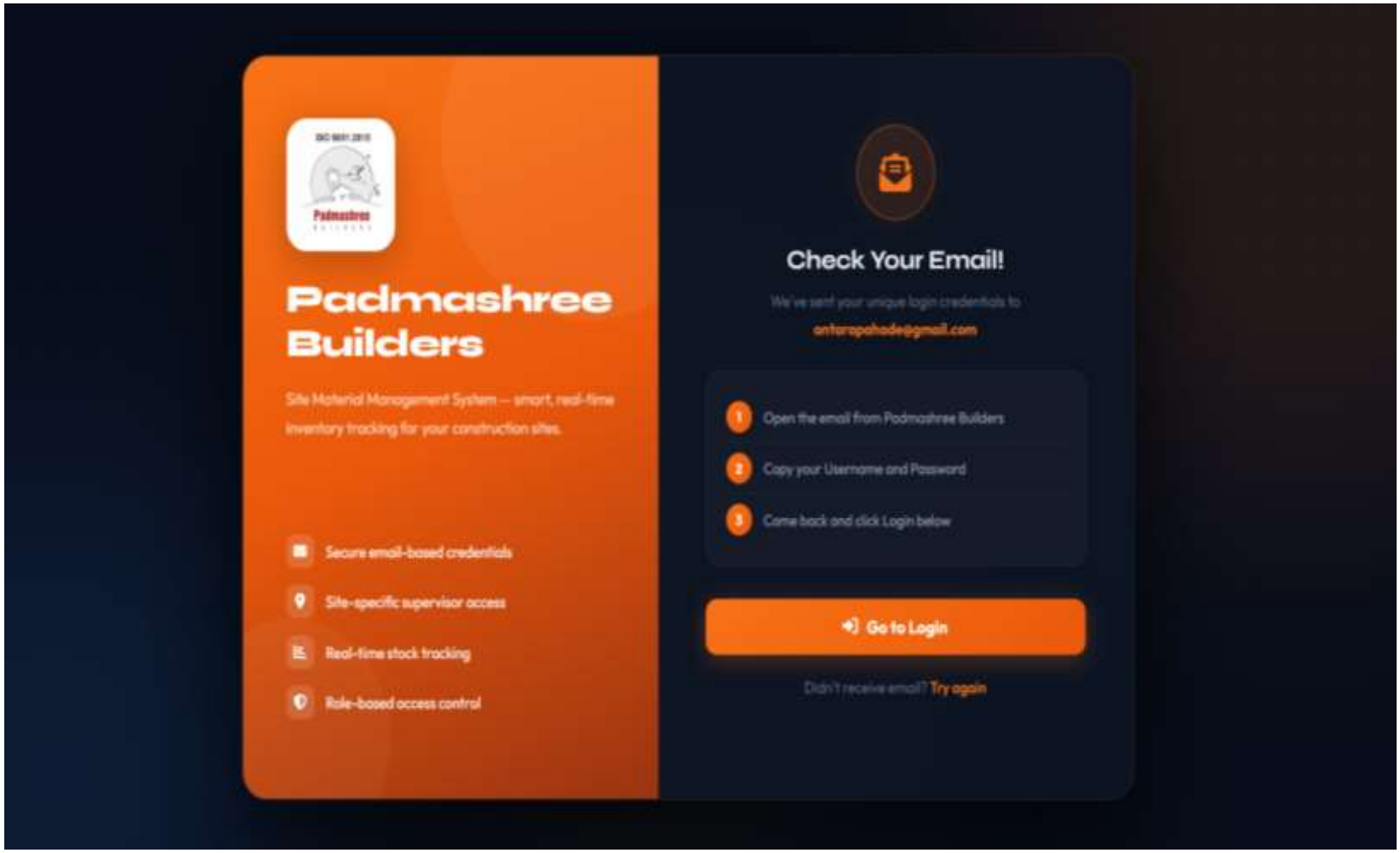


6. Results and Applications

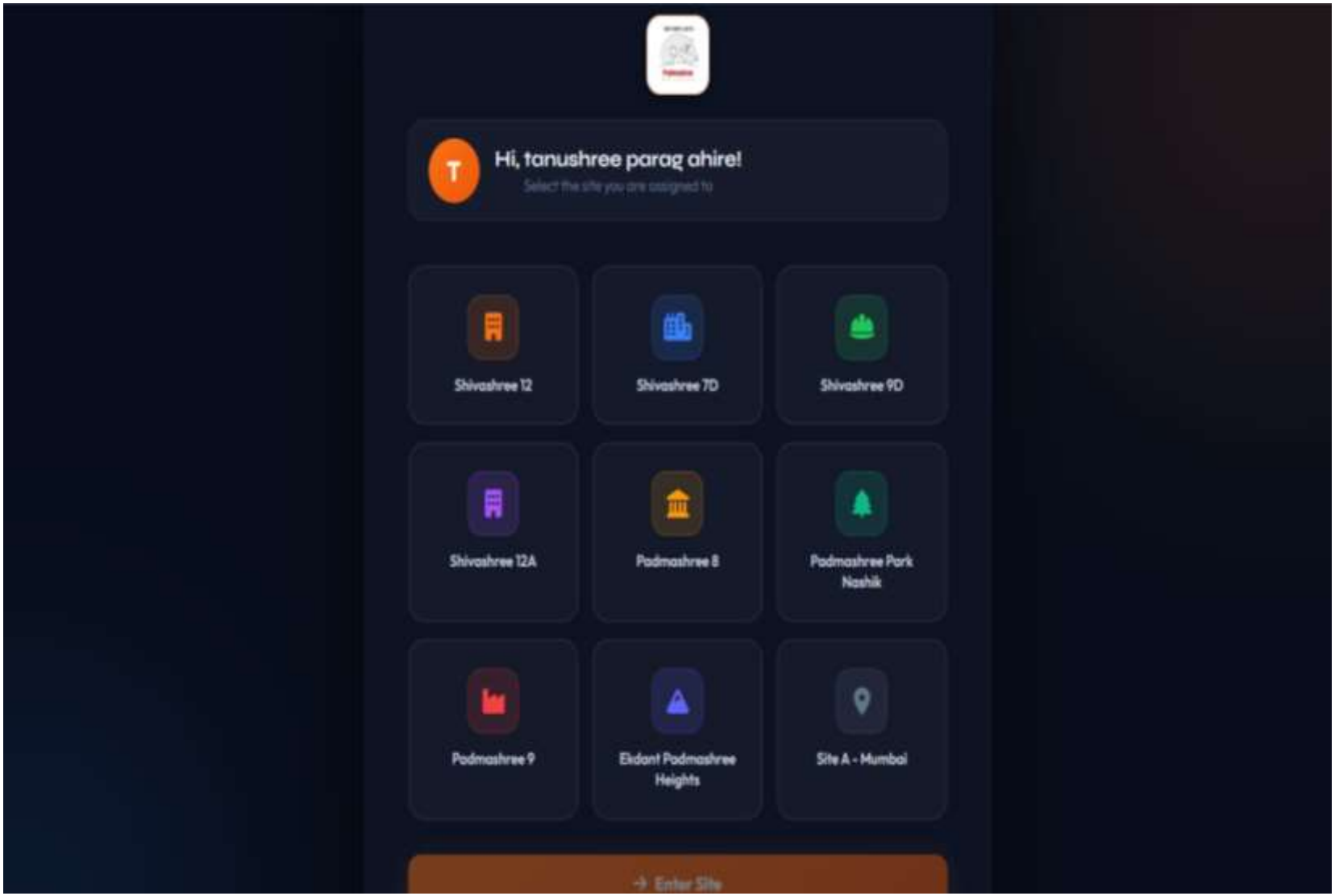
- **Home/Login Page**



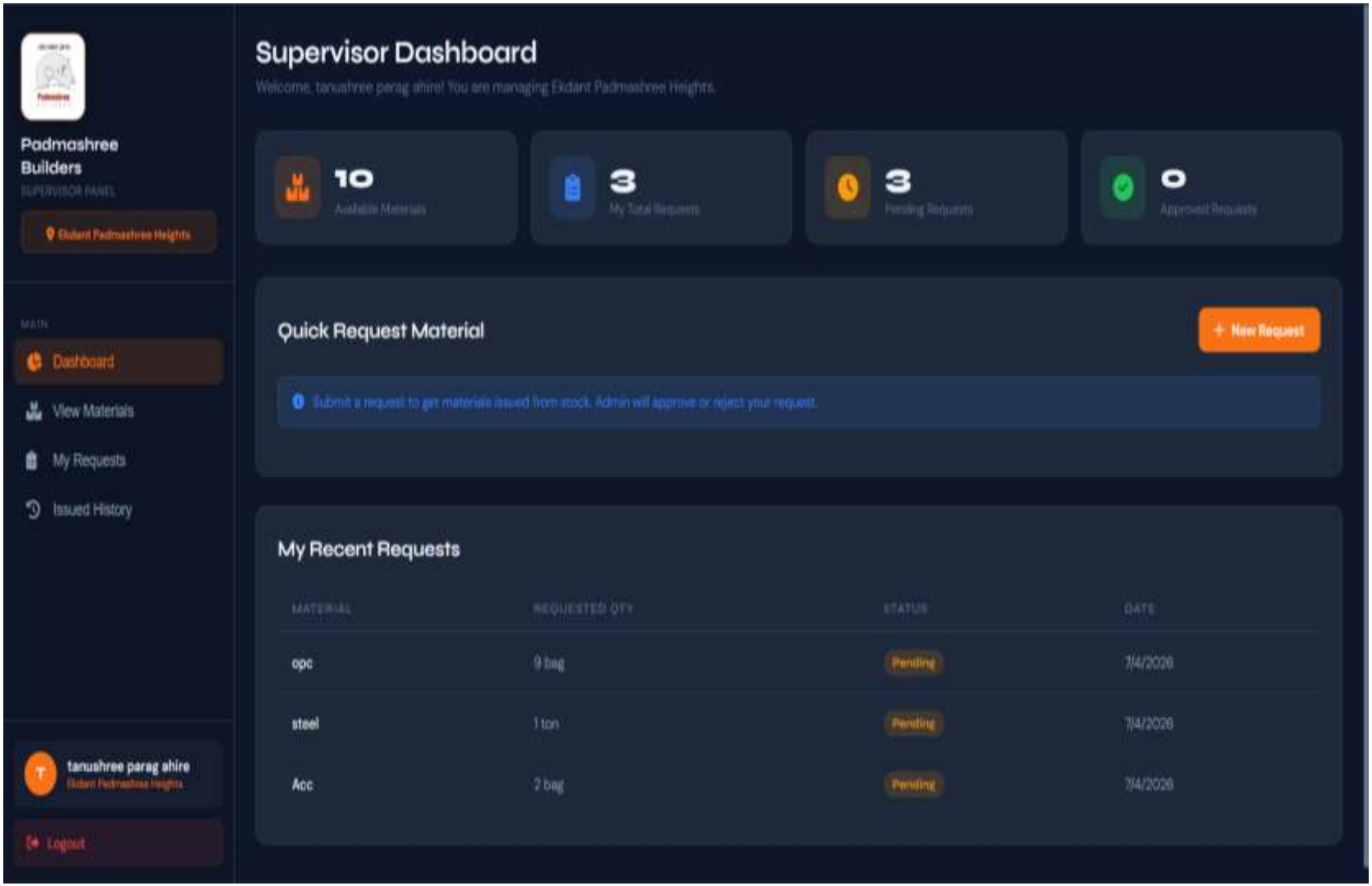
- **Email Page**


Padmashree Builders
Site Material Management System — smart, real-time inventory tracking for your construction sites.
Secure email-based credentials
Site-specific supervisor access
Real-time stock tracking
Role-based access control
Check Your Email!
We've sent your unique login credentials to
antarapahade@gmail.com
1 Open the email from Padmashree Builders
2 Copy your Username and Password
3 Come back and click Login below
Go to Login
Didn't receive email? Try again

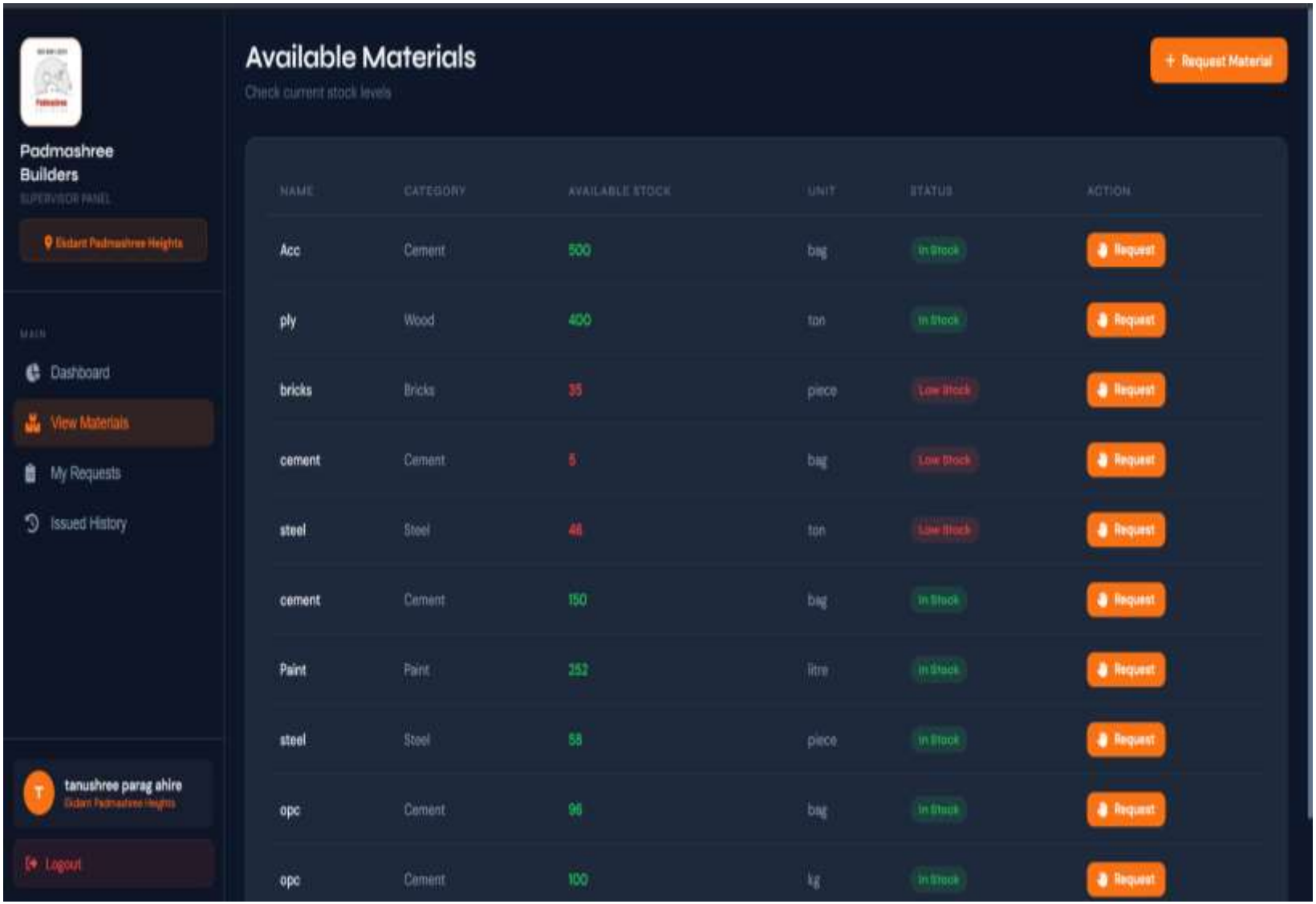
- **Select Site Page**



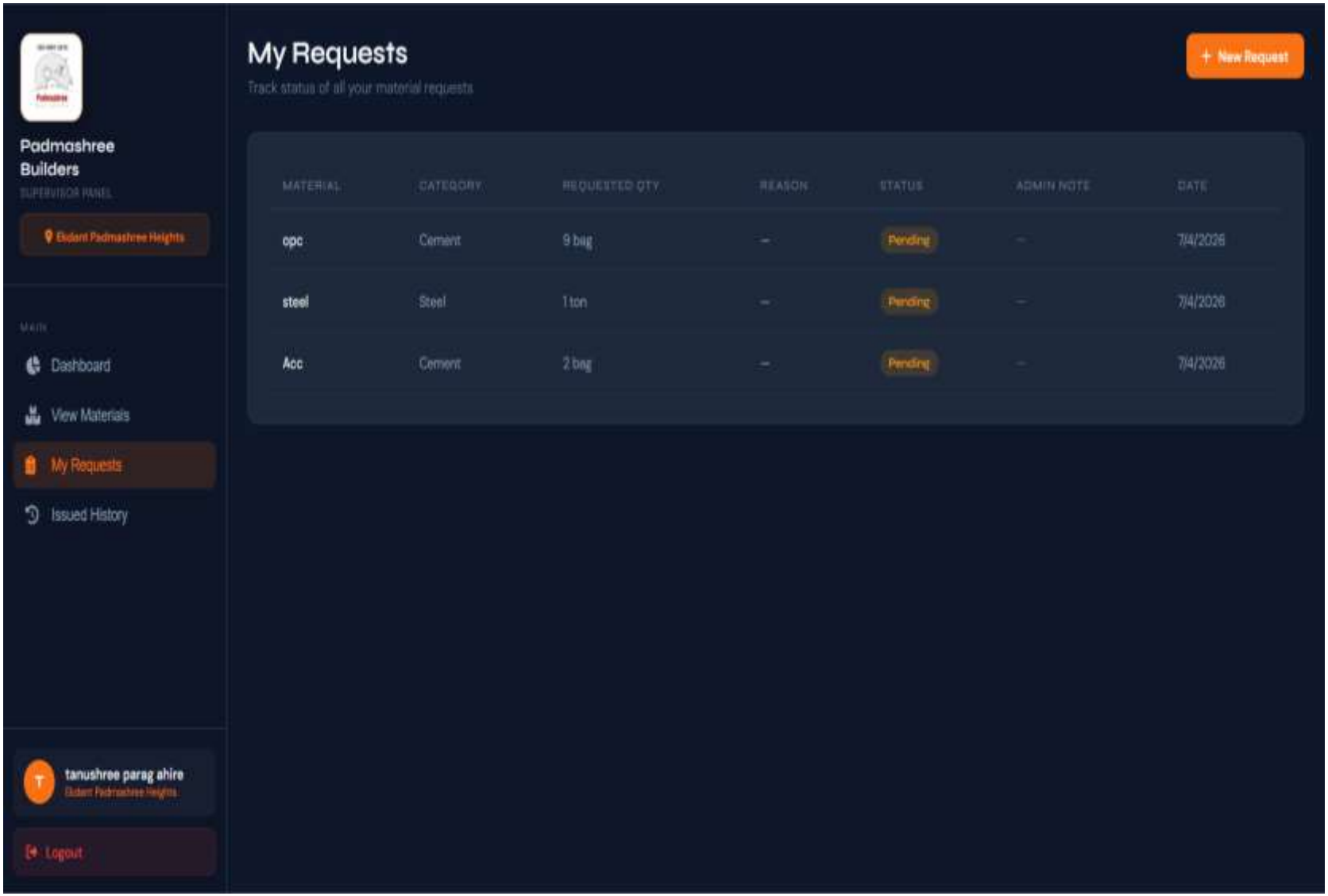
- **Supervisor Dashboard Page**



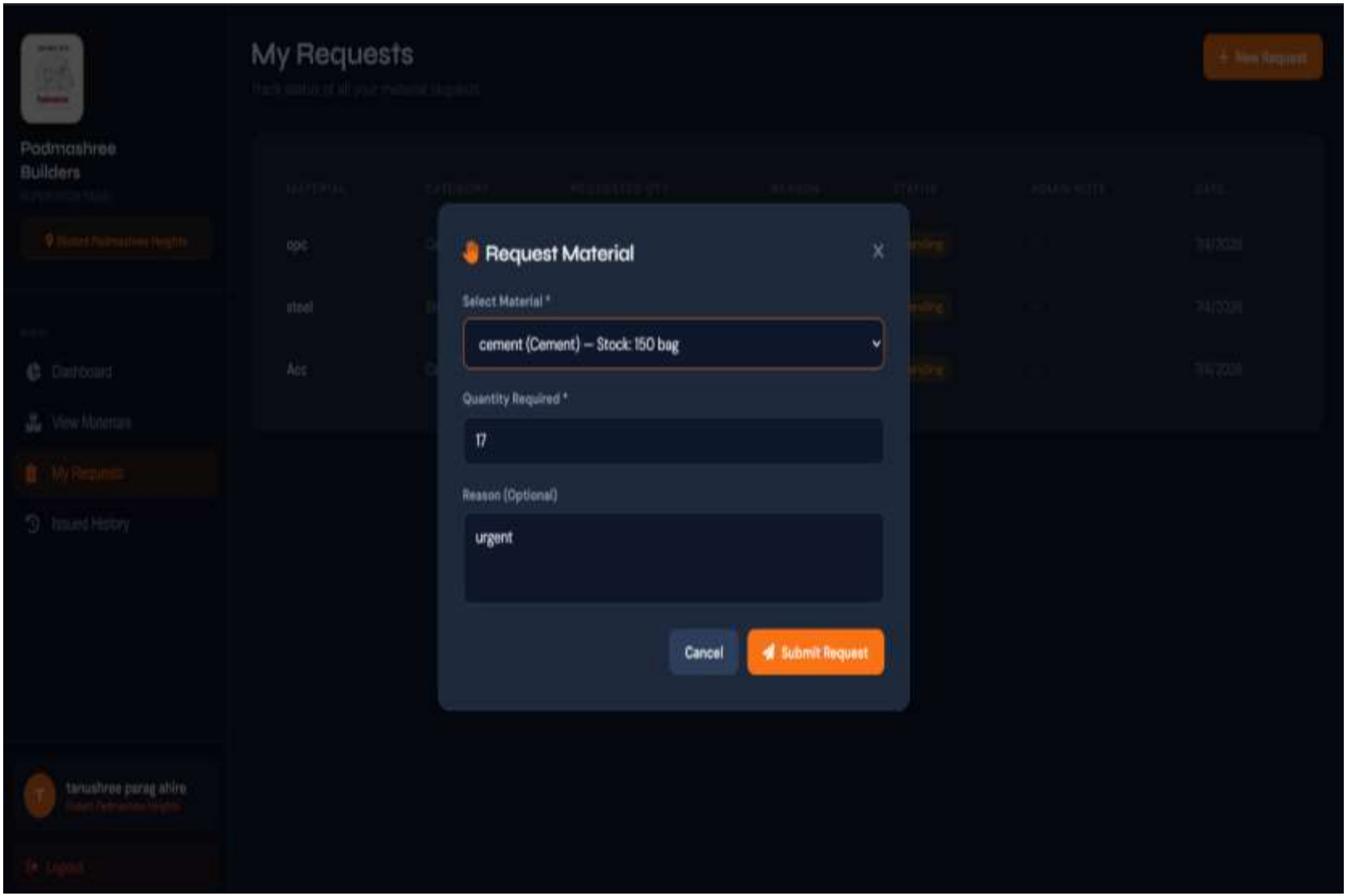
- **View Material Page**



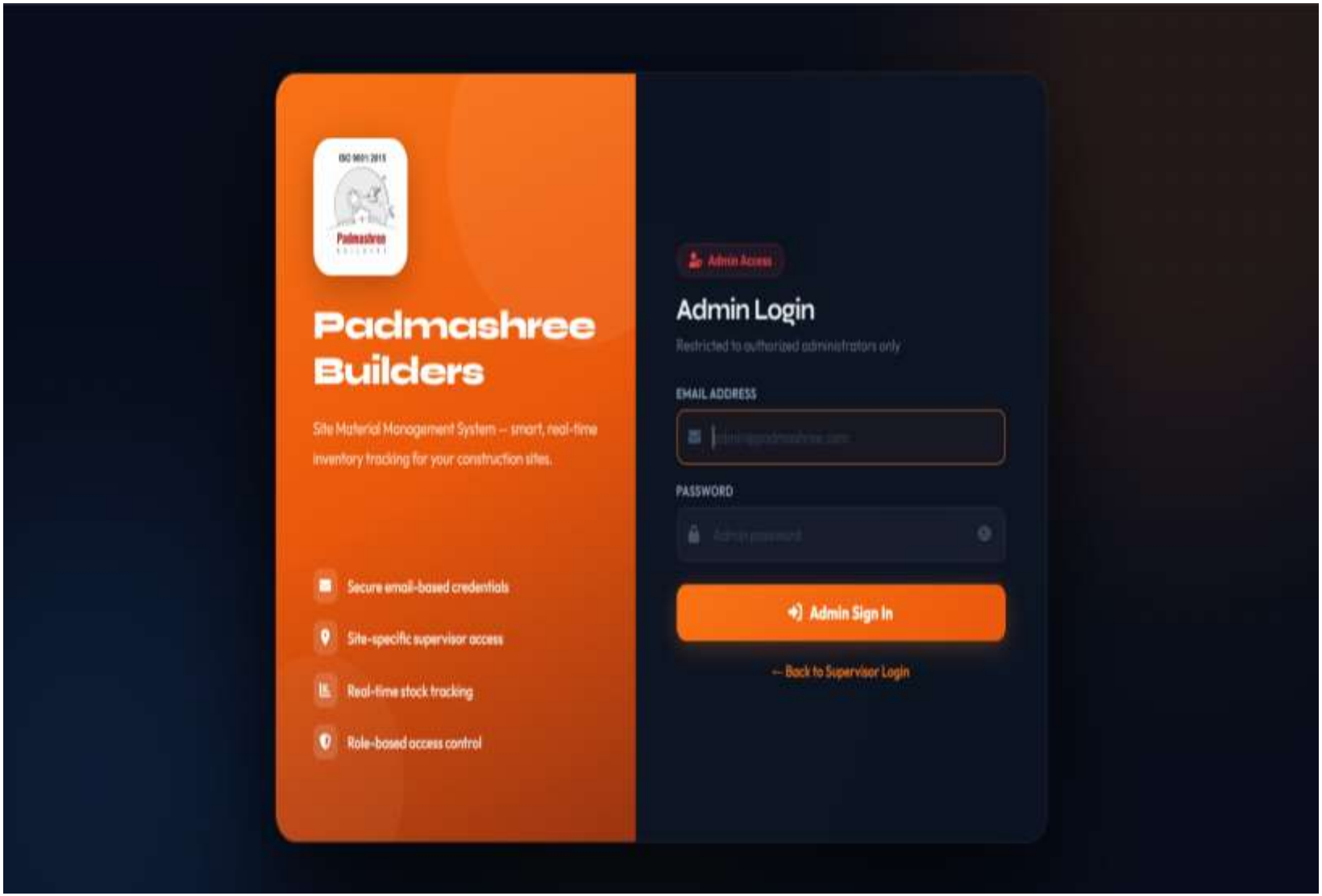
- **My Request Page**



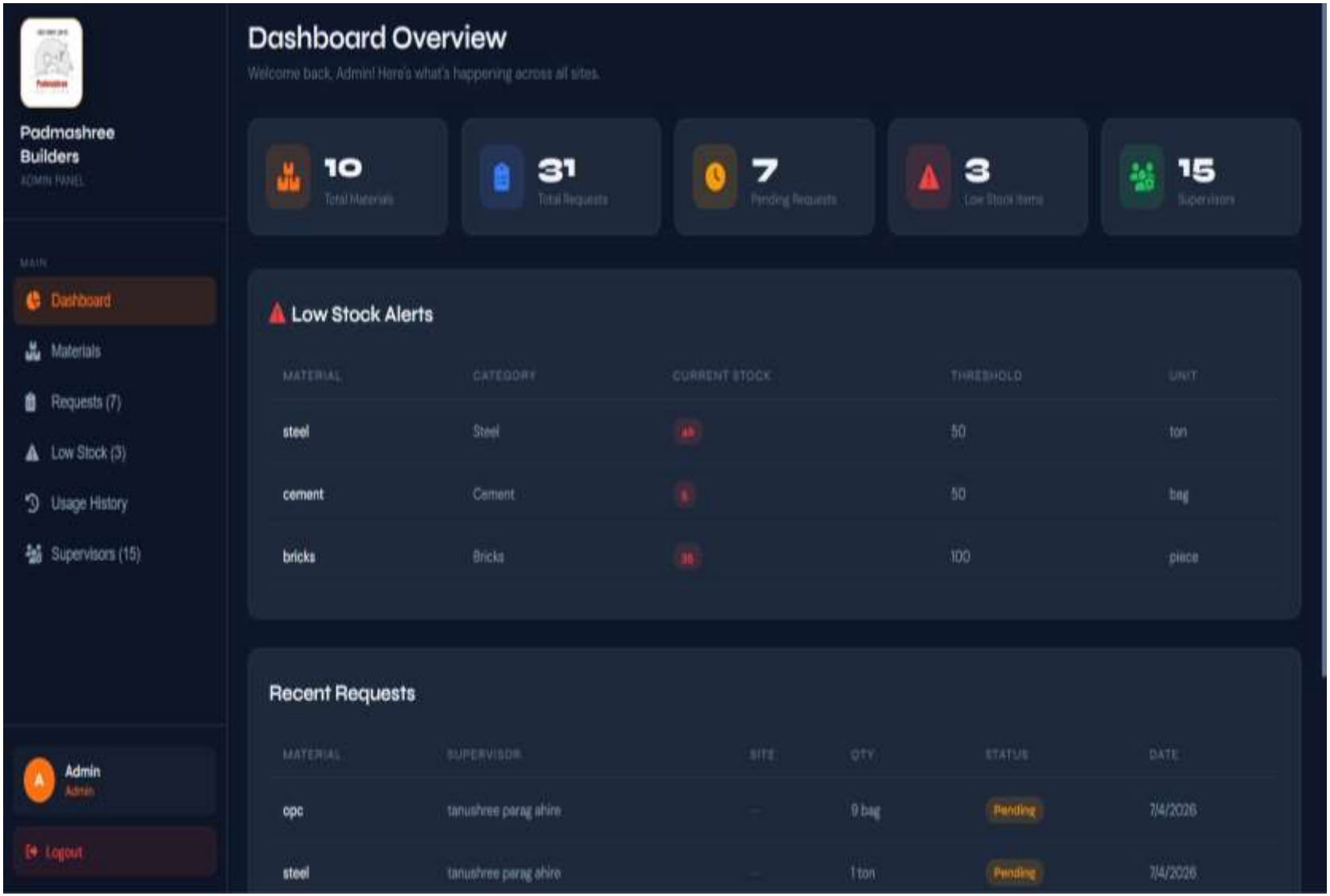
- **My Request Form Page**



- **Admin Login**


Padmashree Builders
Site Material Management System – smart, real-time inventory tracking for your construction sites.
Secure email-based credentials
Site-specific supervisor access
Real-time stock tracking
Role-based access control
Admin Access
Admin Login
Restricted to authorized administrators only
EMAIL ADDRESS
PASSWORD
Admin Sign In
← Back to Supervisor Login

- **Admin Dashboard**



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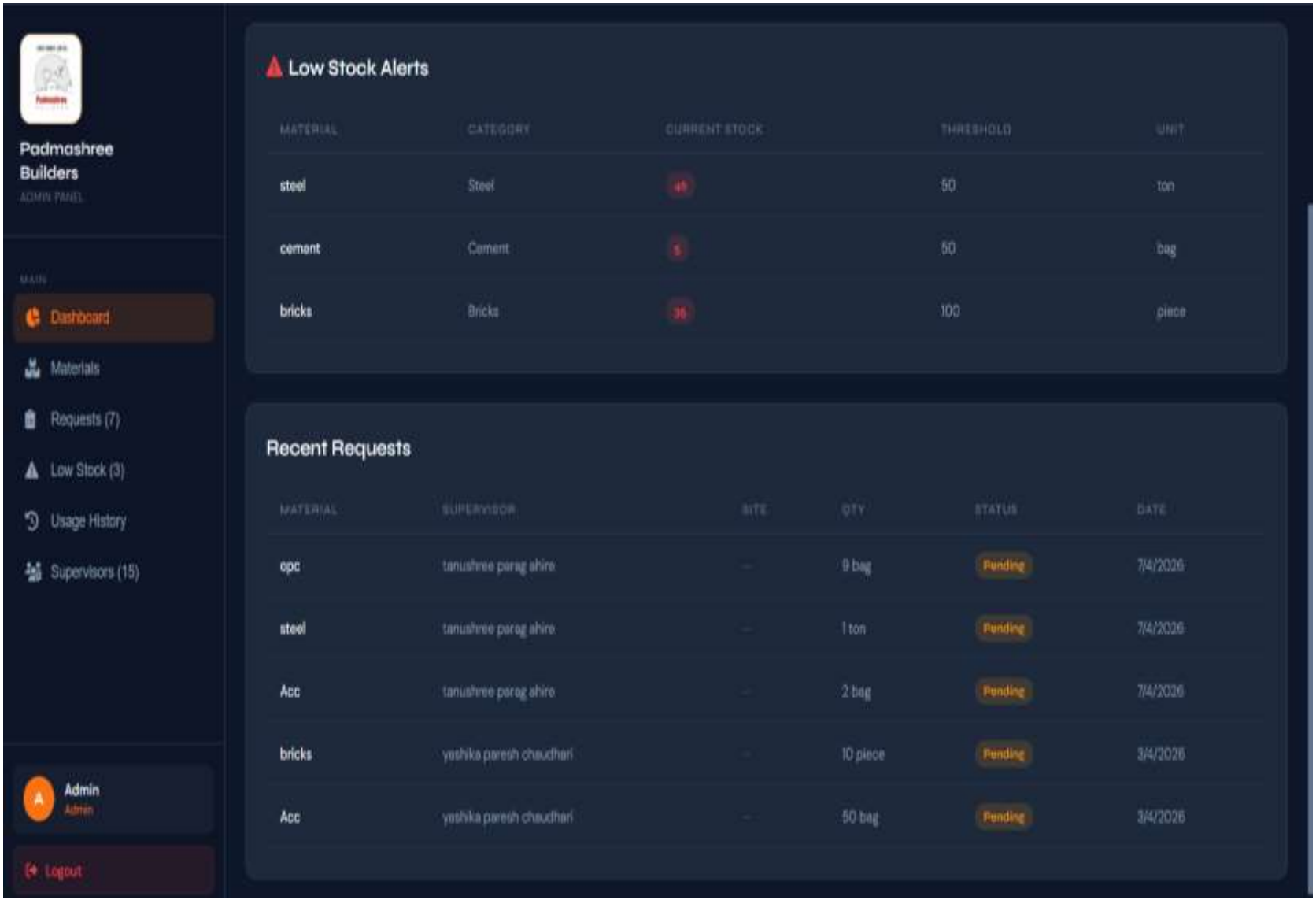
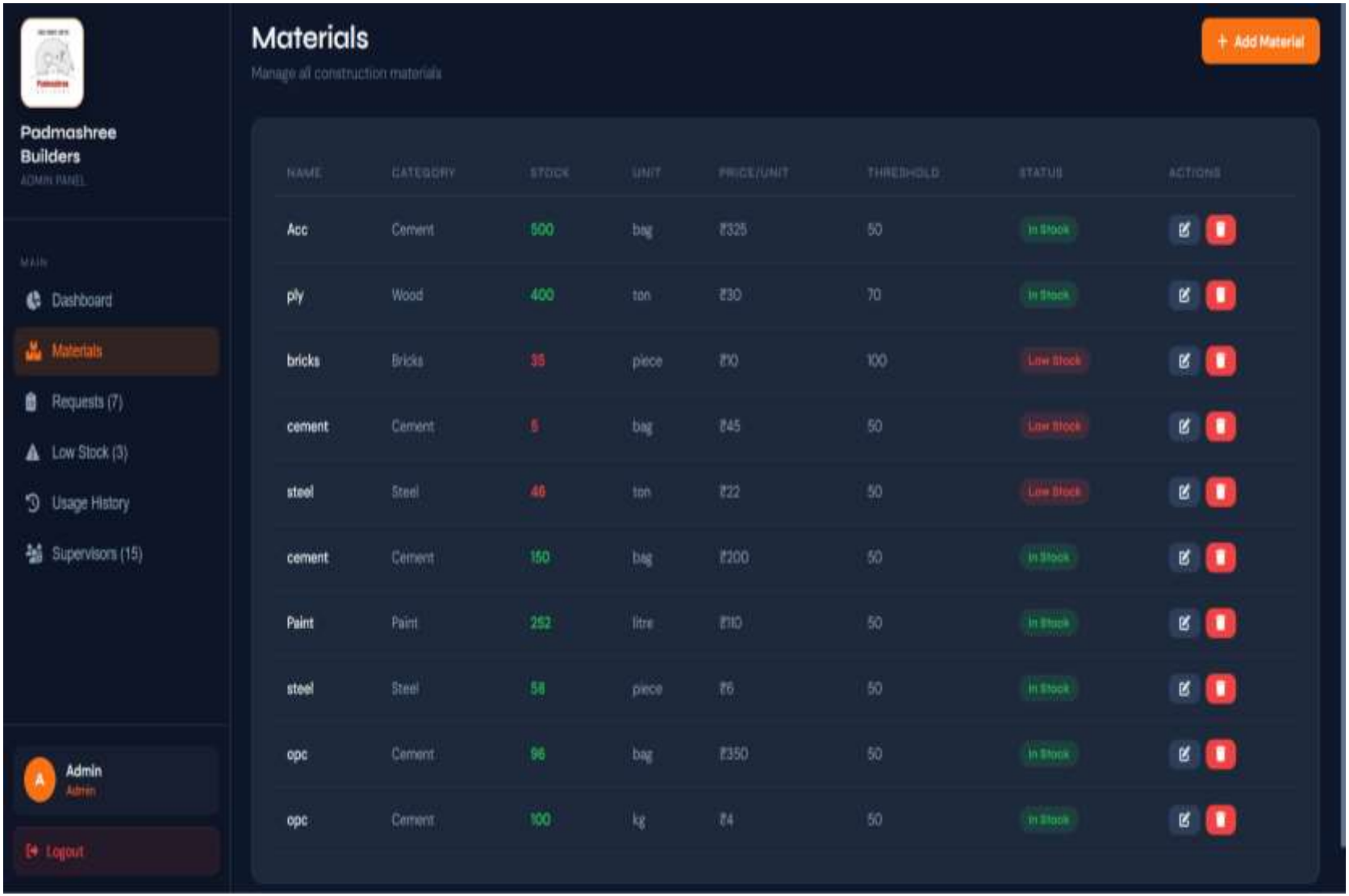
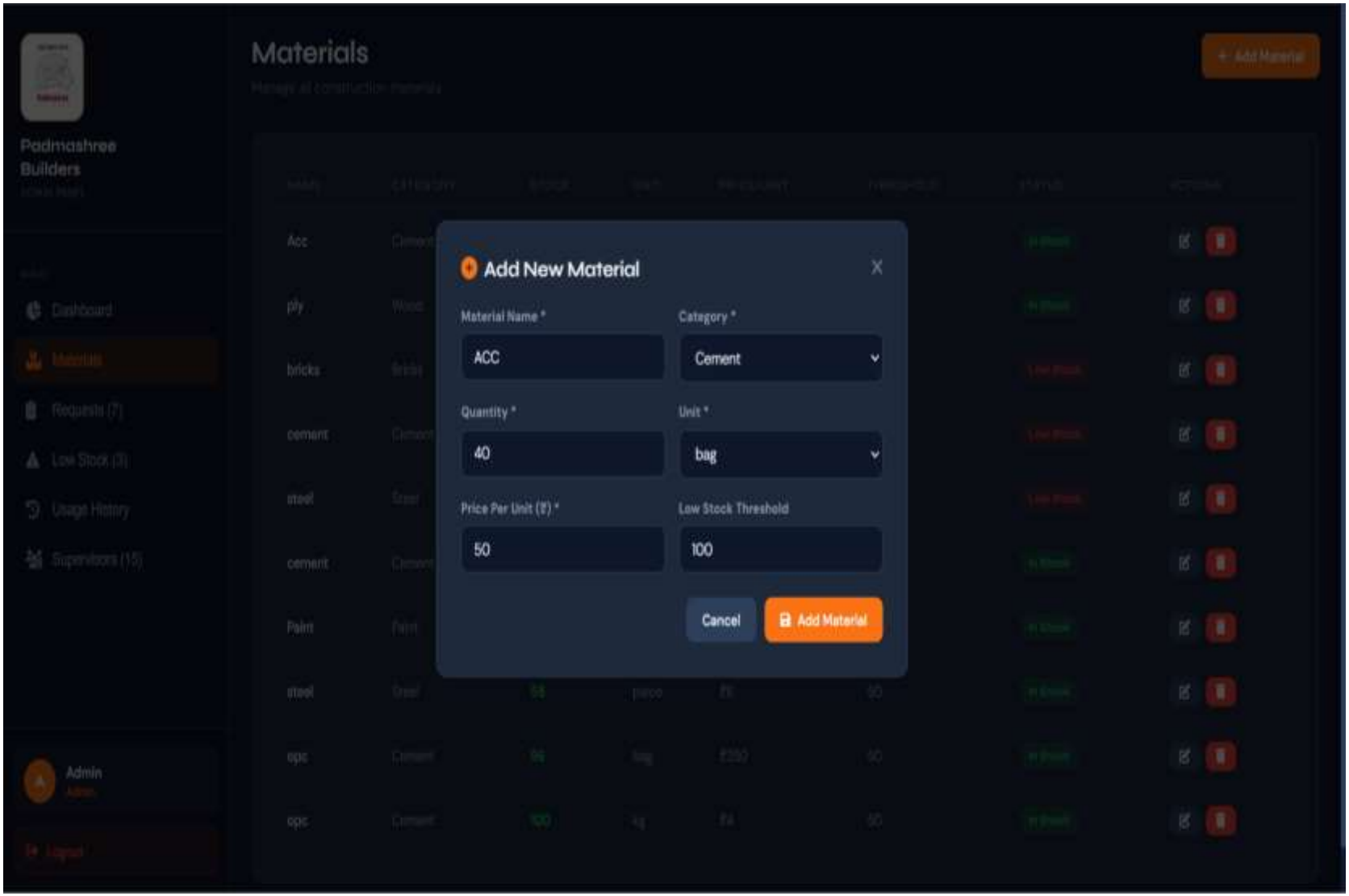


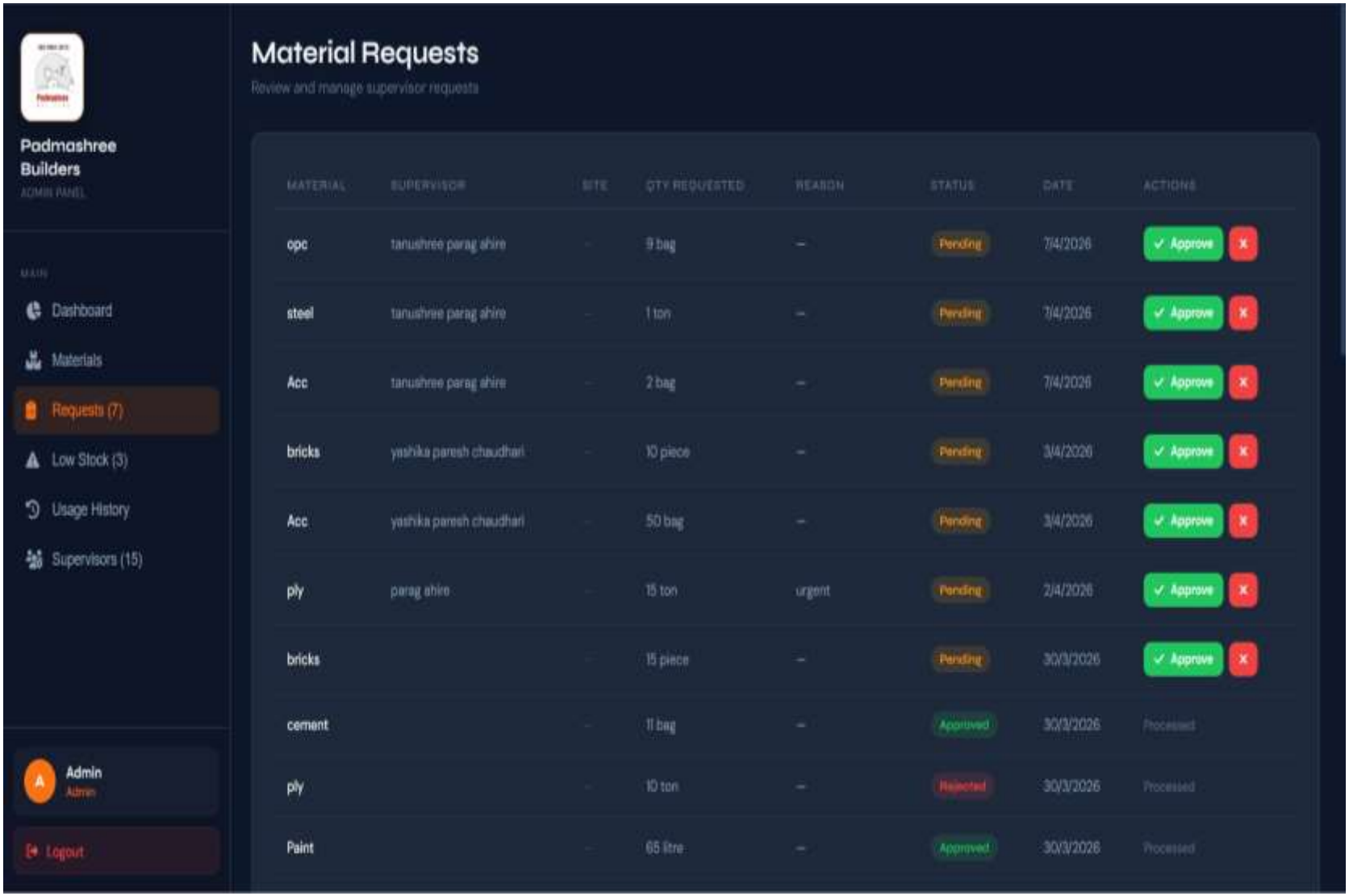
- **Materials Page**



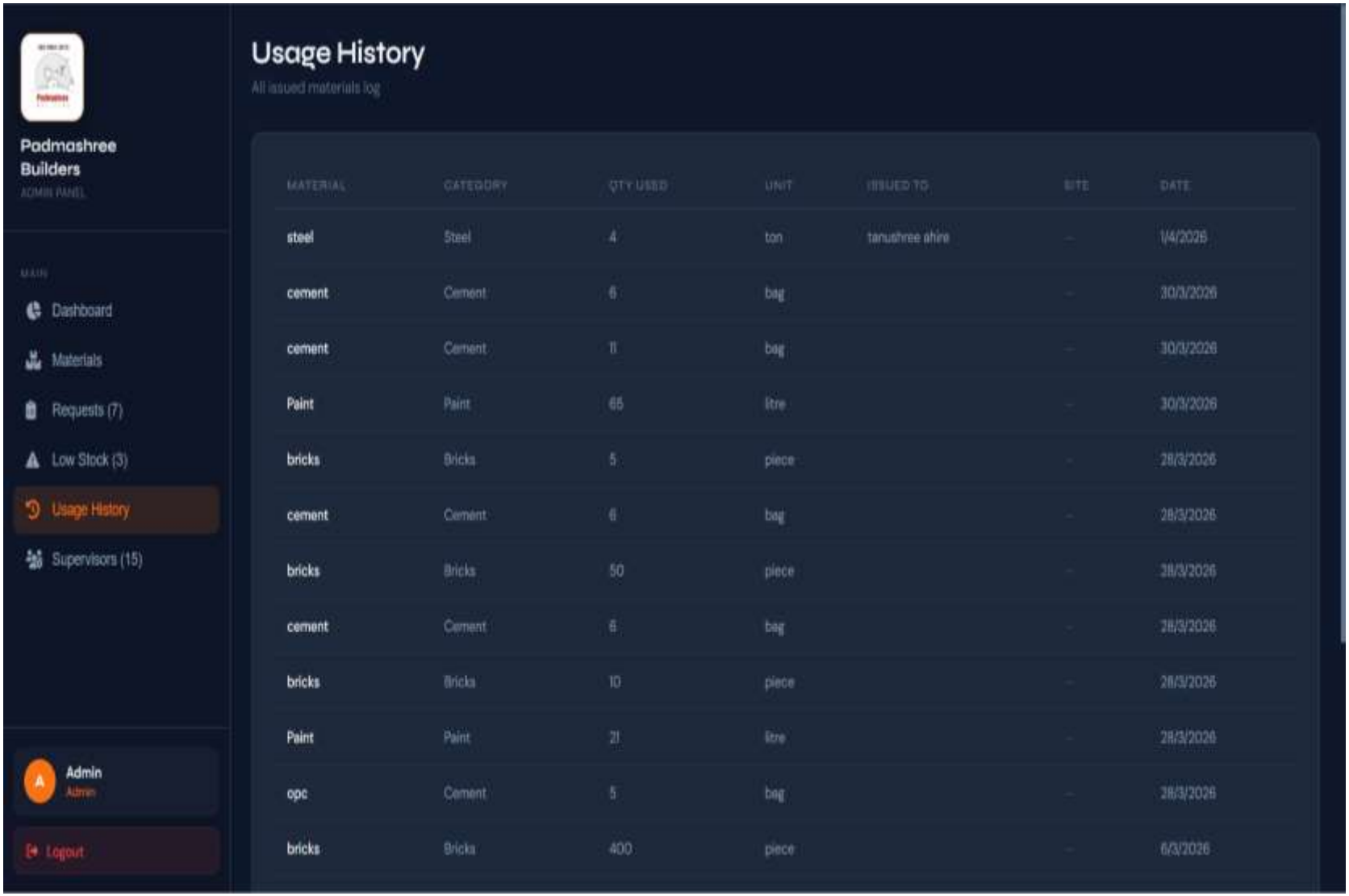
- **Add New Material Form**



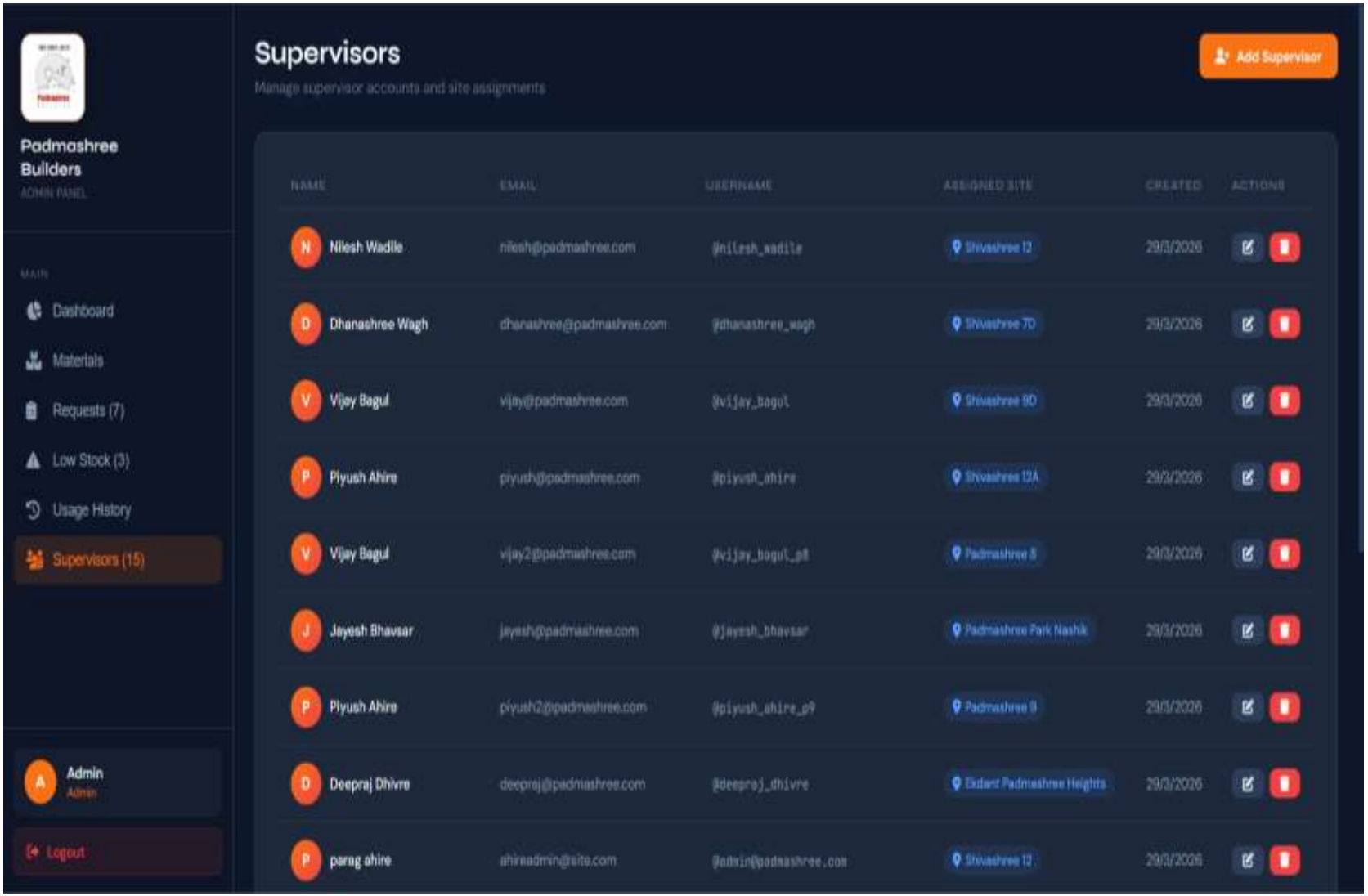
- **Request Material Page**



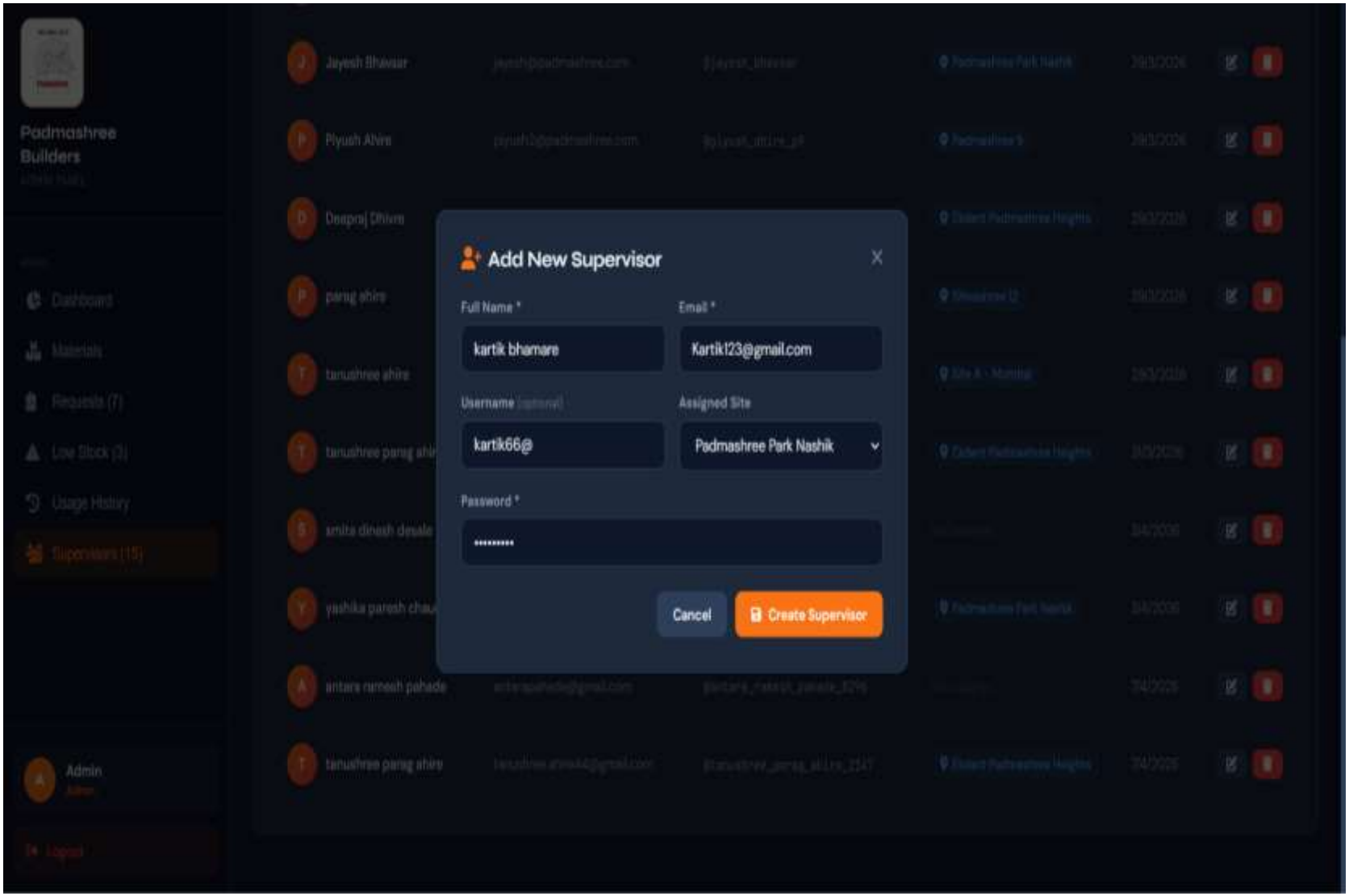
- **Usage History Page**



- **Supervisor Page**



- **Add New Supervisor Form**



7.1 Conclusion

The Site Material Management System simplifies and improves the process of managing construction materials by replacing manual methods with a digital solution. It ensures accurate record keeping, real-time tracking of materials, and efficient handling of requests and approvals. The system enhances transparency, reduces errors, and helps in better decision-making, ultimately improving overall project efficiency

7.2 Future Scope

- Integration with mobile application for easy access on-site
- Implementation of barcode/QR code scanning for material tracking
- Advanced analytics and reporting for better decision-making
- Integration with billing and accounting systems
- Multi-site management with centralized control
- Use of AI for demand prediction and stock optimization

8. References and Bibliography

- JavaScript Documentation
- React JS Official Documentation
- Node.js and Express.js Documentation
- MongoDB Official Documentation
- www.w3schools.com
- www.geeksforgeeks.org